● DSPY 3.4.1 — PYTHONINTERPRETER IMPROVEMENTS, FASTER GEPA, AND MCP V2 COMPATIBILITY · LEARN MORE →

Program, don't prompt, your LLMs.

DSPy is a Python framework for building AI systems. Express your tasks as structured signatures, not prompts, to produce maintainable, modular, and optimizable programs.

```
$ pip install -U dspy
```

Getting Started →

python ≥ 3.10   MIT license   Stanford NLP   github.com/stanfordnlp/dspy

Change LLM   Add a Field   Make it an Agent

extract_events.py

```python
lm = dspy.LM("openai/gpt-5.4-nano")

class ExtractEvent(dspy.Signature):
    """Extract event details from an email."""
    email: str = dspy.InputField()
    event_name: str = dspy.OutputField()
    date: str = dspy.OutputField()

extract = dspy.Predict(ExtractEvent)
extract(email=inbox_message)
```

OUTPUT

```
Prediction(
  event_name="Team Offsite",
  date="Thursday, June 5"
)
```

5.3M+
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IN PRODUCTION AT
databricks shopify Dropbox

Compose programs with reusable primitives.

Signatures

Declare your task.

Define your task as typed inputs and outputs instead of managing messy prompts. Portable, maintainable, and easy to iterate on.

Learn about Signatures →

```python
class Triage(dspy.Signature):
 """Route a support ticket."""
 ticket: str = dspy.InputField()
 urgency: Literal["low", "high"] =
dspy.OutputField()
 team: str = dspy.OutputField()
```

Modules

Same interface, different strategy.

Modules control how your signature executes. Reason, run ensembles, use tools, add a REPL, and more without rewriting your task.

Explore Modules →

```python
# Direct completion
classify = dspy.Predict(Triage)

# Add step-by-step reasoning
classify = dspy.ChainOfThought(Triage)

# Add tools and a reasoning loop
classify = dspy.ReAct(Triage, tools=[search])
```

Optimizers

Compile your program against a metric.

Give DSPy examples and a scoring function. It tunes your prompts automatically until quality converges.

Try Optimizers →

```python
tp = dspy.GEPA(
 metric=semantic_f1,
 auto="medium")
opt = tp.compile(rag, trainset)

# Before: 0.41 F1
# After: 0.63 F1

opt.save("rag.v2.json")
```

Ask AI

Define a task. Grow it into a system.

Ask AI

Extract   Agent   Pipeline   Multimodal   Optimize

```python
class Extract(dspy.Signature):
 """Extract contact info."""
 message: str = dspy.InputField()
 name: str = dspy.OutputField()
 email: Optional[str] = dspy.OutputField()
 intent: Literal[
 "meeting", "intro", "follow-up"
 ] = dspy.OutputField()

extract = dspy.Predict(Extract)
extract(message="I'm Sarah"
 "(sarah@acme.co). Meet Thursday?")
```

OUTPUT   STDOUT

```
Prediction(
 name="Sarah",
 email="sarah@acme.co",
 intent="meeting"
)
```

Signatures define tasks and enforce output types

Learn more about Signatures →

Ask AI

Built in the open, since Dec 2022.

DSPy started at Stanford NLP and grew into a research community. New optimizers and module types land here first — then show up in production systems at companies you've heard of.

Dec 2025	Recursive Language Models	arxiv →
Jul 2025	GEPA: Reflective Prompt Evolution	arxiv →
Jul 2024	BetterTogether: Fine-Tuning + Prompt Opt.	arxiv →
Jun 2024	MIPROv2: Optimizing Instructions & Demos	arxiv →
Feb 2024	STORM: Writing Wikipedia-like Articles	arxiv →
Oct 2023	DSPy: Compiling Declarative LM Calls	arxiv →
Dec 2022	Demonstrate-Search-Predict	arxiv →

DSPy in production

Shopify
Metadata extraction across all shops; ~550× cost reduction

Dropbox
Optimized Dash relevance judge for ranking and evaluation

AWS
Prompt migration from larger to smaller models on Amazon Nova

JetBlue
Multiple chatbot use cases on Databricks

Replit
Code repair pipeline using code LLMs to synthesize diffs

Databricks
LM judges, RAG, classification, and customer solutions

Nous Research
Evolutionary self-improvement for the Hermes agent

More →
See all companies using DSPy in production

Ask AI

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512+
MERGED PRS / YR

60+
TUTORIALS & RECIPES

GitHub →   Discord →

Next
Program, don't prom

Ask AI

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